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TITLE PAGE

- **Problem Statement ID – SIH26001**
- **Problem Statement Title – AI-Based Early Warning and Landslide Risk Monitoring System in NER**
- **Theme** – Disaster Management
- **PS Category** – Software
- **Team ID** – SIH26-BIEW-AIDS-01
- **Team Name – Landslide Guardians**
- **TEAM MEMBERS**
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IDEA TITLE

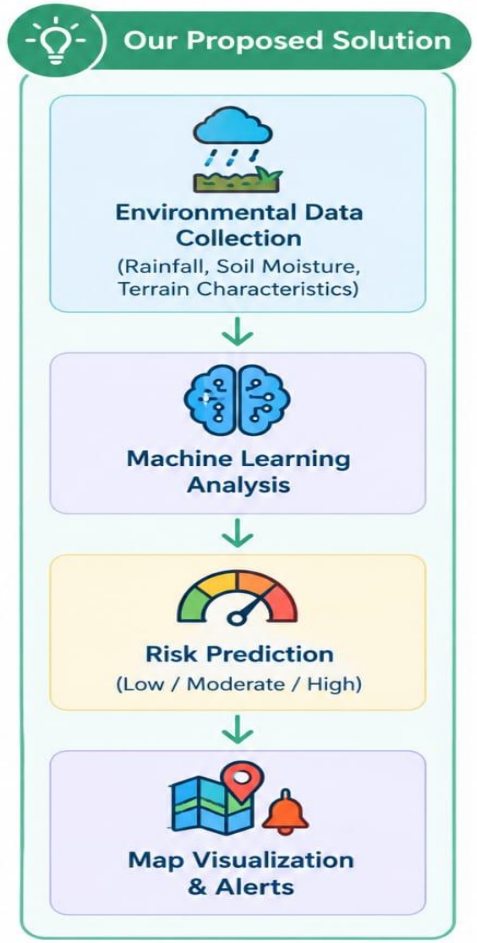
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❖ The Problem

- Landslides are a major natural hazard in the **North Eastern Region (NER)**.
- Heavy rainfall can increase the possibility of landslides.
- Remote and hilly regions are difficult to monitor continuously.
- Existing monitoring may not provide sufficiently early warnings.

❖ Proposed Solution

- Develop an **AI-based automated landslide risk monitoring system**.
- Collect rainfall, soil moisture and terrain-related data.
- Analyze real-time and historical data using **Machine Learning**.
- Generate **Low / Moderate / High risk levels**.
- Display vulnerable locations through a **GIS-based risk map**.
- Send early alerts when the risk reaches a critical level.



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TECHNICAL APPROACH

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Component	Technology
○ Programming	Python
○ Data Processing	Pandas, NumPy
○ Machine Learning	Scikit-learn / XGBoost
○ Database	MySQL / MongoDB
○ Dashboard	HTML, CSS, JavaScript / React
○ Maps	GIS / Map API
○ Sensors	IoT sensors
○ Deployment	Cloud

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FEASIBILITY AND VIABILITY

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❖ Feasibility

- Uses commonly available **AI, ML and GIS technologies**.
- Can initially work with **historical/public datasets**.
- IoT sensors can be added for real-time monitoring.
- Web-based dashboard allows centralized monitoring.

❖ Challenges

- Limited sensor availability in remote regions.
- Poor internet connectivity.
- Missing or noisy data.

❖ Solutions

- Combine multiple data sources.
- Apply data cleaning and validation techniques.
- Use offline/local data storage where required.
- Continuously update and retrain the ML model.

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IMPACT AND BENEFITS

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❖ Target Users

- Disaster Management Authorities
- Local Government Administration
- Emergency Response Teams
- Communities in vulnerable areas
- Road & Infrastructure Authorities

❖ Social Impact

- Improves **early awareness and preparedness**.
- Supports faster response in vulnerable areas.
- Helps communities make timely safety decisions.

❖ Economic Benefits

- Helps reduce damage to infrastructure.
- Reduces disruption to roads and transportation.
- Supports better allocation of monitoring resources.

❖ Environmental Benefits

- Helps identify vulnerable slopes.
- Supports continuous monitoring of high-risk terrain.
- Provides useful information for safer planning.

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RESEARCH AND REFERENCES

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❖ Research Areas

- AI-Based Rainfall-Induced Landslide Prediction in North Eastern India
- A 2026 study explored AI/ML-based landslide early warning using rainfall and soil-moisture data in North Eastern India.
- XGBoost performed well for predicting landslide risk.
- The research supports the use of AI, environmental data and risk maps for early warning systems.

Project Link: https : https://github.com/topics/landslide-detection

❖ References

- **Geological Survey of India (GSI)--**Landslide-related geological and hazard information. gsi.gov.in
- **India Meteorological Department (IMD)--**Rainfall and weather information.mausam.imd.gov.in
- **ISRO / NRSC – Bhuvan--**Geospatial and disaster-related mapping resources. bhuvan.nrsc.gov.in
- **National Disaster Management Authority (NDMA)-**Disaster preparedness and landslide-related guidance. ndma.gov.in